

Capstone Topic

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For my capstone I want to look at how electricity/energy gets bought and sold at the wholesale level in the US. This is the deals between the companies that generate power and the utilities and businesses that buy it, not what people pay on their electric bill. Most of these sales are negotiated privately rather than on a public market, so it's generally hard to see what's going on inside them. I'm interested in what firms are actually doing: who locks into long-term contracts versus buying short-term, where they're sourcing their power, and whether that changes when energy policy shifts or costs get unpredictable, maybe with a closer look at cases like renewable companies or data centers. I am also interested in possibly creating a map of energy contracts to visualize the overall flow. My advisor is **Professor Leila Safavi** in the Economics department, who works on energy and environmental economics. The data is FERC's Electric Quarterly Reports, roughly 4 billion transactions since 2013 recording who sold to whom, at what price, and for how long; it was basically unusable until a project called PUDL cleaned it up this year. The size is the main analytic challenge, since it's far too big to load into a data frame. I am planning on querying it with SQL where it's stored and potentially pulling down aggregated summaries. I don't want to commit past that yet: I know I can't do all of this, so my first step is exploring the data to see what's feasible and narrowing down with Professor Safavi.